

The Adolescent Scar of Early-Life Disaster

José Elías Durán Roa

September 29, 2026

Tables and instrument appendix — draft for discussion

The comparison in words. Every adolescent of the 2024 ELPI wave lived through the 27 February 2010 earthquake as a child; what varies is how hard their municipality was shaken and how old they were when it struck, grouped as under age one (0–11 months), age one (12–23), age two (24–35) and ages three to four (36–59). The estimates answer one question. Take the gap in an outcome at ages 14–18 between two exposure ages, say the children struck at 24–35 months versus those struck at 0–11 months, among the municipalities the earthquake hit; take the same gap among the municipalities it did not hit; the coefficient is the first gap minus the second. We therefore never compare affected against unaffected municipalities directly, and never compare cohorts nationally: municipality fixed effects keep every comparison within a municipality, and cohort fixed effects strip out everything shared by a birth cohort, including its age at the survey. All tables estimate versions of

$$Y_{imc} = \alpha + \sum_a \beta_a (\text{Bin}_a^i \times D_m) + \gamma_a \text{Bin}_a^i + \psi_m + \theta_c + X_i' \delta + \varepsilon_{imc} \quad (1)$$

where i is an adolescent, m the municipality where the household lived before the earthquake, and c the birth cohort. Bin_a marks the child's age, in months, when the earthquake struck. D_m is a dummy equal to one when the municipality belongs to the affected zone, the six regions where the shaking was destructive (V to IX and the Metropolitan Region); in the dose tables it is replaced by the peak ground acceleration, that is, how strongly the ground actually shook in the municipality, expressed as a share of the acceleration of gravity and standardized. ψ_m and θ_c are municipality and cohort fixed effects and X_i are household controls. Standard errors are clustered by municipality and estimates use the 2024 sampling weights.

THE MAIN COEFFICIENT IN WORDS

Children who lived through the earthquake at age two (24–35 months old) reach ages 14–18 with better mental health and less anxiety than children who lived through it before age one (0–11 months) or at ages three to four (36–59 months). In particular, mental health, measured with the PHQ-4, an internationally validated index that groups four questions on low mood, loss of interest, nervousness and uncontrollable worry, is 0.39 points better on its 0-to-12 scale, where the sample mean is 4.31; in standard deviations, 0.147 (column 3 of Table 3). Anxiety, captured by the anxiety half of the same index, shows a probability of screening positive 7.3 percentage points lower, on a base of 27.3 percent. The magnitude is moderate, because it equals about six tenths of the girl–boy gap in positive anxiety screens and matches the 0.17–0.24 standard-deviation scar the same earthquake left on vocabulary (Table 2).

The age profile is a U (Figure 1), consistent with two known channels, disrupted care in infancy and traumatic memory at preschool ages, with age two protected by infantile amnesia. Among girls the estimate doubles, to 0.300 standard deviations (Table 13). The result is flat in a placebo on outcomes fixed at birth, stable across every specification and dose, and holds under permutation inference ($p = 0.045$); adjusting for the family of four outcomes leaves Romano–Wolf p -values at 0.10–0.12, so the full-sample evidence is solid rather than overwhelming and the girls’ result is the sharpest finding.

TABLE 1—THE ANALYSIS SAMPLE BY AGE AT EXPOSURE TO THE EARTHQUAKE

	Age when the earthquake struck				
	<1	1	2	3–4	All
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Sample</i>					
Adolescents	1,419	2,800	2,678	3,106	10,003
In the affected zone (%)	75.8	76.9	76.4	76.7	76.5
Female (%)	48.4	49.6	50.7	49.2	49.6
Age in 2024 (years)	15.2	15.9	16.9	18.0	16.8
Mother’s years of education in 2010	10.46	10.37	10.22	10.18	10.28
<i>Panel B. Mental health at ages 14–18</i>					
PHQ-4 score: sum of four items (0–12)	4.30	4.26	4.39	4.30	4.31
Positive depression screen, PHQ-2 (%)	34.8	34.4	36.0	33.6	34.6
Positive anxiety screen, GAD-2 (%)	28.2	26.5	27.4	27.4	27.3
CBCL internalizing, caregiver (T)	52.6	52.5	52.4	51.9	52.3
<i>Panel C. Retention</i>					
Reinterviewed in 2024 (% of 2010)	61.7	67.6	67.5	69.8	67.3

Notes: Means over the 10,003 adolescents of the 2024 ELPI wave, all born 2006–2009 and therefore exposed to the 27 February 2010 earthquake between 6 and 50 months of age, by age on the day of the earthquake. The All column pools the four exposure-age groups: every adolescent of the wave, struck between 6 and 50 months of age. The PHQ-4 is a four-item screener (two depression, two anxiety items) answered by the adolescent in private, scored 0–12 and standardized within age in years; a positive PHQ-2 (GAD-2) screen is a depression (anxiety) subscore of three or more, and coefficients for these outcomes are probabilities. The affected zone comprises the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Panel C uses the 14,855 children of the 2010 baseline with a valid birth date and shows that retention into 2024 is flat across exposure-age groups (Table 9).

TABLE 2—THE COGNITIVE SCAR PERSISTS INTO ADOLESCENCE: GILLMORE’S EQUATION
AT THE 14-YEAR HORIZON

	(1)	(2)	(3)	(4)
<i>Panel A. Receptive vocabulary: Peabody picture–word test (z)</i>				
Affected × affected zone	−0.226** (0.108)	−0.227** (0.108)	−0.240** (0.104)	−0.412** (0.180)
Observations	13,779	13,779	13,779	13,779
<i>Panel B. Internalizing problems: caregiver CBCL checklist (z)</i>				
Affected × affected zone	0.057 (0.088)	0.059 (0.088)	0.084 (0.088)	−0.066 (0.125)
Observations	11,038	11,038	11,038	11,038
Municipality fixed effects	Yes	Yes	Yes	Yes
Cohort and wave fixed effects	No	Yes	Yes	Yes
Household controls	No	No	Yes	Yes
Municipality linear trends	No	No	No	Yes

Notes: Each panel reports the coefficient on Affected × affected zone from Gillmore’s (2026) equation (1) taken to the 14-year horizon: the affected group are the children exposed in utero to age four, measured in the 2024 wave at ages 14–18; the comparison group are children conceived after the earthquake, measured in the 2017 wave. The CBCL is oriented as in Gillmore (2026), so that positive values mean fewer problems. The affected zone comprises the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Standard errors clustered by municipality in parentheses; evaluation sampling weights. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 3—AGE AT EXPOSURE TO THE EARTHQUAKE AND SELF-REPORTED MENTAL HEALTH AT AGES 14–18

	(1)	(2)	(3)	(4)
<i>Panel A. Mental health: PHQ-4 index, four questions on low mood, loss of interest, nervousness and worry (0–12, z)</i>				
Exposed at age 1	–0.051 (0.080)	–0.044 (0.078)	–0.045 (0.078)	–0.030 (0.079)
Exposed at age 2	–0.144** (0.072)	–0.153** (0.069)	–0.147** (0.069)	–0.120 (0.076)
Exposed at ages 3–4	–0.038 (0.064)	–0.048 (0.060)	–0.044 (0.060)	–0.011 (0.063)
Contrast: ages 3–4 – age 2	0.106* (0.060)	0.105* (0.061)	0.103* (0.060)	0.109* (0.066)
<i>Panel B. The anxiety half of the index: positive GAD-2 screen (its two anxiety questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.014 (0.032)	–0.012 (0.032)	–0.012 (0.033)	–0.018 (0.035)
Exposed at age 2	–0.073** (0.032)	–0.076** (0.032)	–0.073** (0.032)	–0.064* (0.035)
Exposed at ages 3–4	–0.015 (0.027)	–0.018 (0.028)	–0.016 (0.028)	–0.015 (0.030)
Contrast: ages 3–4 – age 2	0.058** (0.027)	0.058** (0.027)	0.058** (0.027)	0.049* (0.029)
<i>Panel C. The depression half of the index: positive PHQ-2 screen (its two depression questions, subscore of 3 or more)</i>				
Exposed at age 1	–0.017 (0.039)	–0.014 (0.038)	–0.014 (0.038)	–0.020 (0.041)
Exposed at age 2	–0.046 (0.042)	–0.050 (0.041)	–0.048 (0.041)	–0.073 (0.045)
Exposed at ages 3–4	–0.008 (0.032)	–0.011 (0.031)	–0.010 (0.031)	–0.017 (0.035)
Contrast: ages 3–4 – age 2	0.038 (0.032)	0.038 (0.033)	0.038 (0.033)	0.056 (0.036)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Observations	9,996	9,996	9,996	6,430
Selection municipalities (clusters)	116	116	116	67

Notes: Ordinary least squares estimates of equation (1) on the adolescents of the 2024 ELPI wave; seven of the 10,003 adolescents of Table 1 drop for incomplete PHQ-4 answers. Panels B and C disaggregate the Panel A index into its two pairs of questions, expressed as clinical screens; Appendix Table A1 lists every question verbatim. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. The affected zone comprises the six regions where the official intensity study records destructive shaking (V, VI, VII, VIII, IX and Metropolitan; Astroza et al. 2010). Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother's education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. The sample is the same in every panel; column 4 has 6,430 adolescents in 67 municipalities. Demanding inference for column 3 is reported in Table 7. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

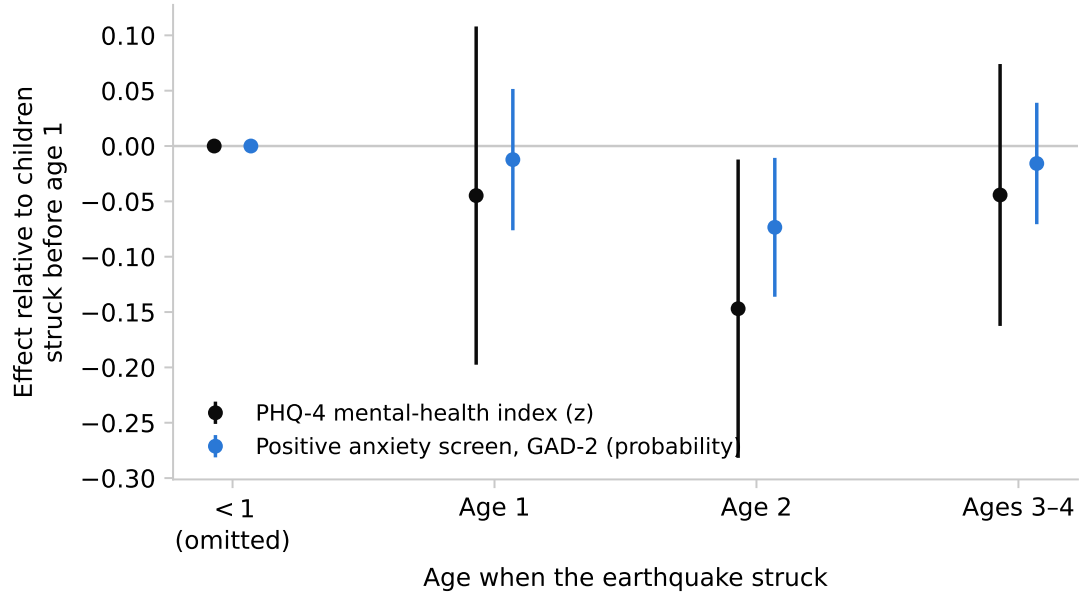


Figure 1. The U-Shaped Gradient of Adolescent Mental Health in the Age at Exposure

Notes: Each marker is a β_a coefficient of equation (1) with the controls of column 3 of Table 3, with its 95 percent confidence interval; the omitted group was struck at 0–11 months. The PHQ-4 is in standard deviations and the GAD-2 screen in probability units. Standard errors clustered by municipality of selection in parentheses.

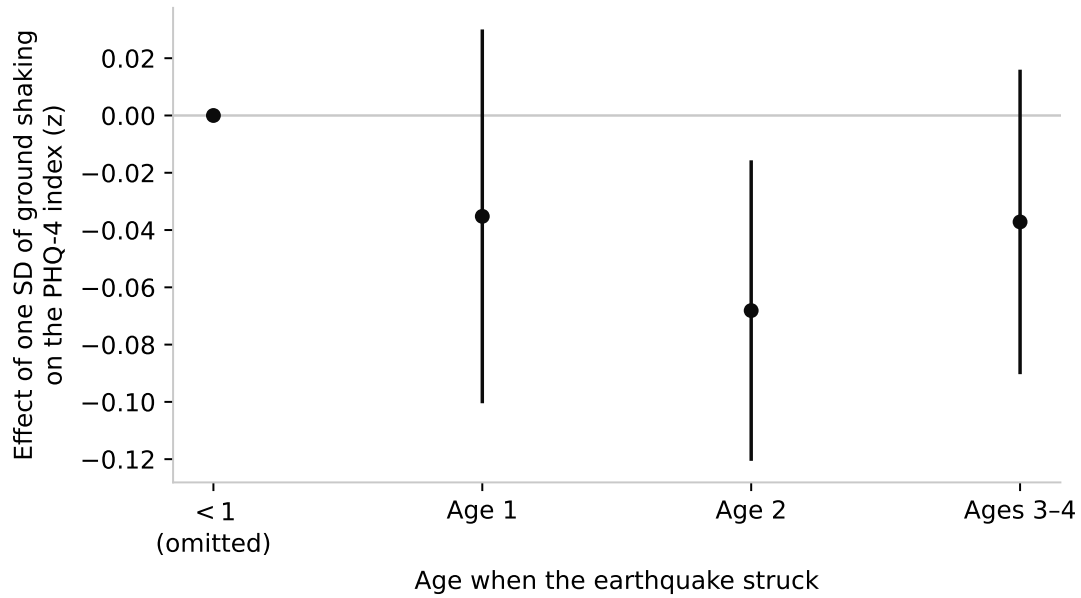


Figure 2. The Gradient Under the Continuous Seismic Dose

Notes: Each marker is the coefficient on the exposure-age group interacted with the standardized peak ground acceleration, how strongly the ground shook, at the centroid of the municipality of selection, with its 95 percent confidence interval; specification of column 3 of Table 3. The dose varies across the 116 municipalities, within affected regions. Standard errors clustered by municipality of selection in parentheses.

TABLE 4—THE CAREGIVER DOES NOT SEE IT: PARENT-REPORTED OUTCOMES

	(1)	(2)	(3)	(4)
<i>Panel A. CBCL internalizing score in 2024: caregiver checklist (T, z)</i>				
Exposed at age 1	−0.057 (0.099)	−0.047 (0.096)	−0.055 (0.095)	−0.024 (0.100)
Exposed at age 2	−0.121 (0.074)	−0.121 (0.075)	−0.109 (0.075)	−0.131 (0.084)
Exposed at ages 3–4	−0.086 (0.069)	−0.084 (0.068)	−0.077 (0.069)	−0.069 (0.074)
Observations	9,970	9,970	9,970	6,412
<i>Panel B. Within-child change in the same CBCL score, 2017 to 2024 (z by wave)</i>				
Exposed at age 1	−0.030 (0.131)	−0.011 (0.125)	−0.012 (0.124)	−0.020 (0.128)
Exposed at age 2	−0.009 (0.104)	−0.004 (0.102)	−0.002 (0.102)	−0.045 (0.110)
Exposed at ages 3–4	−0.040 (0.109)	−0.049 (0.106)	−0.050 (0.106)	−0.093 (0.112)
Observations	7,169	7,169	7,169	4,901
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	No	Yes	Yes	Yes
2012 controls	No	No	Yes	Yes
Excludes the Metropolitan Region	No	No	No	Yes
Selection municipalities (clusters)	116	116	116	67

Notes: Same design and specifications as Table 3, with the caregiver’s report of the child as the outcome. The CBCL internalizing score is reported by the main caregiver, in T units, standardized within age. Panel B uses the change between the 2017 measurement (ages 7–11) and the 2024 measurement (ages 14–18) of the same child, each standardized within its wave and age; with two periods this is equivalent to a child fixed-effects estimate. Neither panel shows an exposure-age gradient: the scar of Table 3 appears only in what adolescents report in private. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Column 1 includes exposure-age, municipality-of-selection and birth-cohort fixed effects; column 2 adds pre-earthquake controls (sex, age in months, and the 2010 household baseline: mother’s education and age, household size, rurality); column 3 adds 2012 controls (number of children of the mother, family history of mental illness), with missing-value indicators; column 4 estimates column 3 without the Metropolitan Region. Estimates use the 2024 sampling weights. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 5—THE GRADIENT UNDER CONTINUOUS SEISMIC DOSES

	Seismic dose of the municipality of selection			
	Affected zone (0/1)	PGA (z)	MMI (z)	–log dist. to epicenter (z)
	(1)	(2)	(3)	(4)
<i>Panel A. PHQ-4 score: sum of four depression–anxiety items (z)</i>				
Exposed at age 1	–0.045 (0.078)	–0.035 (0.033)	–0.028 (0.032)	–0.029 (0.029)
Exposed at age 2	–0.147** (0.069)	–0.068** (0.027)	–0.062*** (0.023)	–0.062** (0.025)
Exposed at ages 3–4	–0.044 (0.060)	–0.037 (0.027)	–0.039 (0.024)	–0.025 (0.023)
<i>Panel B. Positive GAD-2 screen: two anxiety items, subscore of 3 or more</i>				
Exposed at age 1	–0.012 (0.033)	–0.015 (0.014)	–0.013 (0.013)	–0.022* (0.013)
Exposed at age 2	–0.073** (0.032)	–0.039*** (0.015)	–0.029** (0.013)	–0.036*** (0.013)
Exposed at ages 3–4	–0.016 (0.028)	–0.016 (0.013)	–0.010 (0.010)	–0.014 (0.011)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes
Observations	9,996	9,996	9,996	9,996
Selection municipalities (clusters)	116	116	116	116

Notes: Estimates of equation (1) under four measures of the seismic dose, all with the controls of column 3 of Table 3. Column 1 repeats the official binary affected zone. Column 2 uses the peak ground acceleration: the strongest shaking of the ground during the earthquake, expressed as a share of the acceleration of gravity, measured at the centroid of the municipality of selection (from the ShakeMap of the U.S. Geological Survey) and standardized; column 3 an intensity scale from the same source; column 4 minus the log of the distance from the centroid to the epicenter. The continuous doses vary across the 116 municipalities, within affected regions, which addresses inference concerns with a regional treatment. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

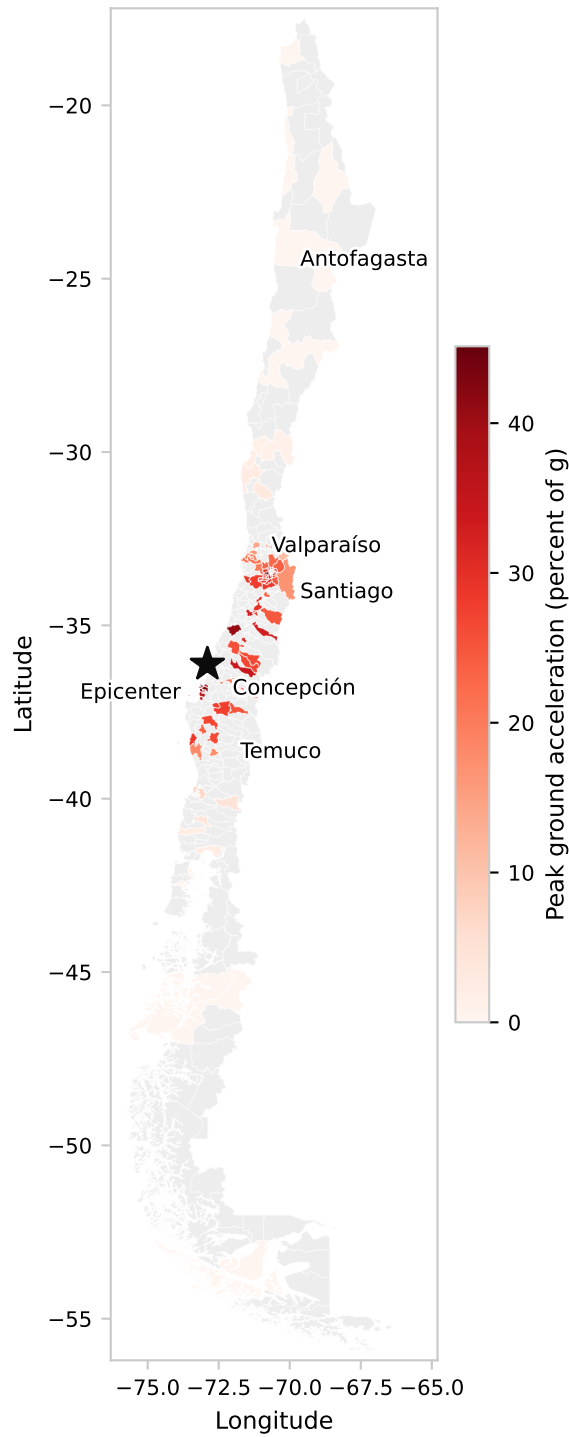


Figure 3. The Seismic Dose Across the Municipalities of Selection

Notes: Each colored municipality is one of the 116 municipalities of selection of the ELPI sample, colored by the peak ground acceleration, that is, how strongly the ground shook during the earthquake, expressed as a share of the acceleration of gravity (from the ShakeMap of the U.S. Geological Survey); the star marks the epicenter and light-gray municipalities are not in the sample. The map shows the two sources of identifying variation of Table 5, the affected zone and the continuous dose within it.

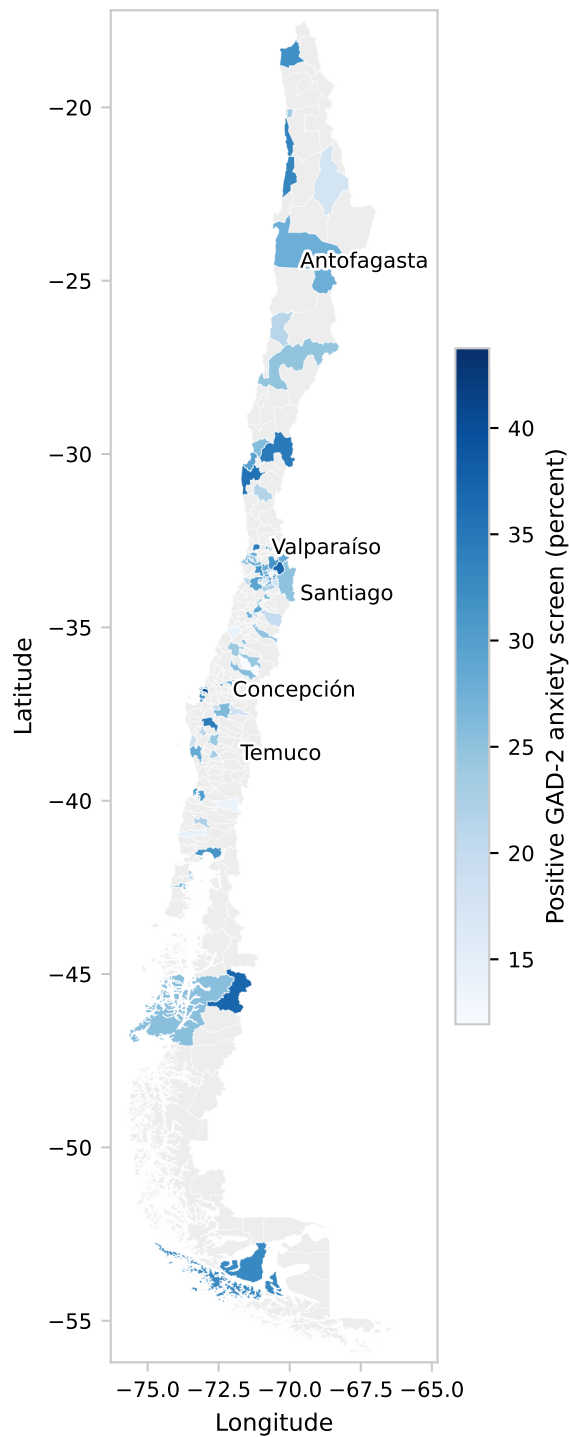


Figure 4. The Geography of Adolescent Anxiety in 2024

Notes: Each colored municipality is one of the 116 municipalities of selection, colored by the share of its adolescents with a positive GAD-2 anxiety screen in the 2024 wave; municipality shares range from 11.9 to 43.8 percent, and light-gray municipalities are not in the sample. The map is descriptive and unadjusted; the regressions of the paper compare cohorts within municipality, never municipalities against each other.

TABLE 6—PLACEBO: OUTCOMES FIXED AT BIRTH, BEFORE THE EARTHQUAKE

	Birth weight (z) (1)	Birth length (z) (2)	Gestation weeks (z) (3)	Premature (0/1) (4)	Apgar 5' (z) (5)
Exposed at age 1	−0.013 (0.067)	−0.029 (0.074)	−0.023 (0.079)	0.017 (0.021)	−0.083 (0.087)
Exposed at age 2	−0.082 (0.080)	−0.074 (0.074)	−0.166** (0.069)	0.041* (0.024)	−0.043 (0.089)
Exposed at ages 3–4	0.017 (0.075)	0.014 (0.060)	−0.001 (0.077)	0.003 (0.021)	−0.013 (0.112)
Municipality fixed effects	Yes	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes	Yes
Observations	9,153	9,104	9,715	9,988	4,143

Notes: Each column reports one estimate of equation (1) on an outcome fixed at birth, one to four years *before* the earthquake for every cohort in the sample, as reported by the mother in 2010 (Appendix Table A5). The design should find nothing. Of fifteen coefficients, two are significant at the 10 percent level (1.5 expected by chance), and their sign — a worse birth endowment of the age-2 group in affected municipalities — would work *against* the trough of Table 3; the birth-endowment row of Table 7 confirms the result does not move. Exposure-age groups give the child's age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 7—ROBUSTNESS AND DEMANDING INFERENCE FOR THE MAIN RESULT

	Dependent variable: PHQ-4 score (z)		Observations
	Age 2 × affected	Ages 3–4 – age 2	
	(1)	(2)	(3)
Preferred specification (column 3)	−0.147** (0.069)	0.103* (0.060)	9,996
Fixed effects only	−0.144** (0.072)	0.106* (0.060)	9,996
Excluding the Metropolitan Region	−0.120 (0.076)	0.109* (0.066)	6,430
Unweighted	−0.129* (0.071)	0.097 (0.063)	9,996
PHQ-4 in points (0–12), not standardized	−0.391** (0.183)	0.277* (0.162)	9,996
Balanced sample with a 2017 CBCL	−0.114 (0.082)	0.083 (0.075)	7,188
Controlling for the birth endowment	−0.144** (0.069)	0.100* (0.060)	9,996
Rewighted for attrition (IPW)	−0.149** (0.069)	0.104* (0.060)	9,996
Continuous dose: PGA (z) instead of the zone	−0.068** (0.027)	0.031 (0.026)	9,996
Dropping one affected region at a time (range)	[−0.172, −0.120]	[0.080, 0.114]	
Municipality and birth-cohort fixed effects (all rows)	Yes		
<i>Demanding inference for the preferred specification (p-values)</i>			
Permutation of the affected zone across municipalities	0.045	0.107	
Wild cluster bootstrap by region (15 regions)	0.132	0.069	

Notes: Each row re-estimates the preferred specification of Table 3, Panel A, changing one thing. Column 1 reports the coefficient on exposure at age two $\times$ affected zone and column 2 the contrast between ages 3–4 and age two. The IPW row reweights by the inverse of the estimated probability of remaining in the 2024 wave (Table 9). The permutation test reassigns the affected-zone status across the 116 municipalities of selection, keeping the number of treated municipalities, 2,000 times. The wild cluster bootstrap by region uses Webb weights and clusters at the level of the regional treatment. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 8—MULTIPLE-HYPOTHESIS ADJUSTMENT ACROSS MENTAL-HEALTH OUTCOMES

	Age 2 $\times$ affected		Ages 3–4 – age 2	
	Unadjusted p	Romano–Wolf p	Unadjusted p	Romano–Wolf p
	(1)	(2)	(3)	(4)
PHQ-4 mental-health index (z)	0.033	0.120	0.088	0.235
Its depression half: PHQ-2 positive	0.247	0.299	0.244	0.468
Its anxiety half: GAD-2 positive	0.022	0.102	0.032	0.111
CBCL internalizing, caregiver (z)	0.147	0.299	0.583	0.615

Notes: Romano–Wolf stepdown p -values over the family of four mental-health outcomes, estimated with the preferred specification of Table 3. The adjustment uses a joint wild cluster bootstrap by municipality (Rademacher weights, 2,000 replications, the same sign flips for every outcome), separately for the age-two coefficient and for the contrast between ages 3–4 and age two.

TABLE 9—ATTRITION FROM 2010 TO 2024 IS NOT DIFFERENTIAL BY EXPOSURE AGE AND ZONE

	Pr(reinterviewed in 2024) (1)
Exposed at age 1 $\times$ affected	0.038 (0.029)
Exposed at age 2 $\times$ affected	0.028 (0.029)
Exposed at ages 3–4 $\times$ affected	0.038 (0.029)
Exposure-age and region fixed effects	Yes
Mother’s education and age in 2010	Yes
Observations	14,855
Mean retention into 2024	67.3 percent

Notes: Linear probability model over the children of the 2010 baseline with a valid birth date. The outcome is an indicator for being reinterviewed in the 2024 wave. The small and insignificant coefficients show that the loss of the sample (32.7 percent) does not concentrate in any treatment-by-age cell; the fitted probabilities feed the IPW row of Table 7. The affected zone enters through the region of residence reported in 2010. Robust standard errors in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 10—OTHER ADOLESCENT OUTCOMES I: WELL-BEING AND VICTIMIZATION

	Resilience (z) (1)	Life satisf. (z) (2)	Poor health (z) (3)	Bullying (z) (4)	Cyber-vict. (0/1) (5)
Exposed at age 1	0.096 (0.076)	0.086* (0.050)	−0.086 (0.066)	0.028 (0.060)	−0.015 (0.029)
Exposed at age 2	0.037 (0.080)	0.098* (0.056)	−0.072 (0.077)	−0.047 (0.071)	−0.007 (0.023)
Exposed at ages 3–4	0.058 (0.070)	0.116* (0.062)	−0.054 (0.070)	0.037 (0.063)	−0.005 (0.025)
Municipality fixed effects	Yes	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes	Yes
Observations	9,996	9,996	9,996	9,176	9,985

Notes: Each column reports one estimate of equation (1) with the specification of column 3 of Table 3 on another outcome of the 2024 wave. Resilience is the mean of the six Brief Resilience Scale items; life satisfaction a single 1–7 item; poor health a single reversed item; bullying the mean of eight frequency items, one reverse-coded; cyber-victimization one if any of six items is answered yes. Appendix Table A3 lists every item and scale verbatim. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 11—OTHER ADOLESCENT OUTCOMES II: DATING VIOLENCE AND SUBSTANCE USE

	Dating viol. (0/1) (1)	Tobacco (0/1) (2)	Alcohol (0/1) (3)	Cannabis (0/1) (4)
Exposed at age 1	0.025 (0.019)	0.012 (0.016)	−0.038 (0.032)	0.018 (0.017)
Exposed at age 2	0.005 (0.020)	−0.002 (0.015)	−0.037 (0.040)	0.014 (0.016)
Exposed at ages 3–4	0.028 (0.021)	−0.022 (0.024)	−0.078** (0.031)	0.013 (0.019)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Pre-earthquake controls	Yes	Yes	Yes	Yes
2012 controls	Yes	Yes	Yes	Yes
Observations	7,101	9,995	9,995	9,995

Notes: Each column reports one estimate of equation (1) with the specification of column 3 of Table 3 on another outcome of the 2024 wave. Dating violence is one if any of three items is answered yes, among adolescents who ever dated; tobacco, alcohol and cannabis refer to the last 12 months, with alcohol one if any of four beverage items is answered yes. Appendix Table A3 lists every item and scale verbatim. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 12—THE GRADIENT IS NOT EXPLAINED BY THE MOTHER’S PRE-EARTHQUAKE MENTAL HEALTH

	PHQ-4 score (z)		Positive GAD-2 screen	
	Vulnerable mother (1)	Other mothers (2)	Vulnerable mother (3)	Other mothers (4)
Exposed at age 1	−0.036 (0.174)	−0.021 (0.080)	−0.022 (0.066)	−0.001 (0.036)
Exposed at age 2	−0.101 (0.129)	−0.124 (0.089)	−0.061 (0.063)	−0.062 (0.040)
Exposed at ages 3–4	−0.003 (0.127)	−0.017 (0.071)	−0.052 (0.063)	0.010 (0.031)
Municipality fixed effects	Yes	Yes	Yes	Yes
Birth-cohort fixed effects	Yes	Yes	Yes	Yes
Observations	2,023	7,973	2,023	7,973

Notes: Split-sample estimates of the preferred specification of Table 3. A vulnerable mother reported, in 2010, a diagnosis of depression, anxiety or post-traumatic stress during the pregnancy of the child, a psychological or psychiatric referral, or diagnosed postpartum depression — all predating the earthquake by years (Appendix Table A4). The contrast is similar in both groups for the PHQ-4 and not smaller among children of non-vulnerable mothers for the GAD-2: the gradient is not a recomposition of pre-existing family mental-health burden. Exposure-age groups give the child’s age when the earthquake struck, defined in months of age as 0–11 (before age 1, the omitted group), 12–23 (age 1), 24–35 (age 2) and 36–59 (ages 3–4). Each coefficient compares the gap between the age group in the row and the omitted group in affected municipalities with the same gap in unaffected municipalities. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

TABLE 13—HETEROGENEITY OF THE MAIN RESULT

	Dependent variable: PHQ-4 score (z)		Observations
	Age 2 × affected	Ages 3–4 – age 2	
	(1)	(2)	(3)
Girls	–0.300*** (0.100)	0.182** (0.086)	4,961
Boys	–0.016 (0.098)	0.042 (0.086)	5,035
Mother with secondary education or less	–0.107 (0.074)	0.071 (0.065)	7,755
Mother with tertiary education	–0.201 (0.188)	0.182 (0.149)	2,053
Urban household in 2010	–0.139* (0.076)	0.129** (0.064)	9,005
Rural household in 2010	–0.238 (0.172)	–0.213 (0.257)	991
Municipality and birth-cohort fixed effects	Yes		

Notes: Each pair of rows estimates the preferred specification of Table 3, Panel A, on the subsample named in the row. The baseline level of symptoms is much higher among girls — 33.9 versus 19.5 percent screen positive for moderate or severe symptoms in the full wave — so similar coefficients imply larger relative effects for boys. Standard errors clustered by municipality of selection in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

APPENDIX. INSTRUMENTS: EVERY ITEM AND ITS SCALE

TABLE A1—MENTAL-HEALTH INSTRUMENTS: ITEMS AND SCALES

	Question, verbatim (1)	Response scale (2)
<i>Panel A. PHQ-4, answered by the adolescent in private (2024 wave)</i>		
d4_1	¿Te has sentido bajoneado(a), deprimido(a), irritable o desesperanzado(a)?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_2	¿Has sentido poco interés o placer al hacer las cosas?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_3	¿Te has sentido muy nervioso(a), angustiado(a) o con los nervios de punta?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
d4_4	¿No has podido dejar de preocuparte o controlar la preocupación?	0 = Nunca; 1 = Algunos días; 2 = Más de la mitad de los días; 3 = Casi todos los días
<i>Panel B. Caregiver-reported behavior</i>		
cbc12	Child Behavior Checklist, internalizing scale, answered by the main caregiver about the adolescent; the survey releases the computed T score, not the copyrighted items.	T score, standardized in the analysis

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. The PHQ-4 score used in the paper is the sum of the four items rescaled to 0–3 each (0–12 in total) and standardized within age in years. The positive PHQ-2 (depression) and GAD-2 (anxiety) screens are the survey-provided indicators for a subscore of three or more on the corresponding pair of items.

TABLE A2—WELL-BEING OUTCOMES: ITEMS AND SCALES

	Question, verbatim (1)	Response scale (2)
<i>Panel A. Resilience: Brief Resilience Scale (items 2, 4 and 6 reverse-coded)</i>		
d2_1	Tiendo a recuperarme rápidamente después de haber vivido situaciones difíciles...	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_2	Tengo facilidad para superar eventos estresantes	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_3	No me toma mucho tiempo recuperarme de un evento estresante	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_4	Es fácil para mí recuperarme cuando algo malo sucede	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_5	Usualmente supero situaciones difíciles con poca dificultad	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
d2_6	Me toma poco tiempo recuperarme de las adversidades en mi vida	1 = Muy en desacuerdo; 2 = En desacuerdo; 3 = Neutral; 4 = De acuerdo; 5 = Muy de acuerdo
<i>Panel B. Life satisfaction and self-rated health</i>		
d3_1	Pensando en una escala de 1 a 7, donde 1 es nada satisfecho(a) y 7 es muy satisfecho(a), ¿qué tan satisfecho(a) estás con tu vida familiar?	1 = Nada Satisfecho(a); 2 = -; 3 = -; 4 = -; 5 = -; 6 = -; 7 = Muy Satisfecho(a)
d1	Considerando tu salud física, dirías que esta es:	1 = Muy mala; 2 = Mala; 3 = Regular; 4 = Buena; 5 = Muy buena

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. The resilience score is the mean of the six items after reverse-coding items 2, 4 and 6, standardized; life satisfaction and self-rated health are standardized, the latter with its sign reversed in the analysis so that higher values mean worse health.

TABLE A3—VICTIMIZATION AND RISK-BEHAVIOR OUTCOMES: ITEMS AND SCALES

	Question, verbatim (1)	Response scale (2)
<i>Panel A. School bullying (frequency scale; standardized mean; item g1_3 reverse-coded)</i>		
g1_1	Mis compañeros(as) se burlan de mí, me dicen sobrenombres	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_2	Me siento solo(a) en el curso	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_3	Lo paso bien con mis compañeros(as) de clase	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_4	Mis compañeros(as) son muy agresivos(as)	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_5	Mis compañeros(as) pelean mucho	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_6	A mis compañeros(as) les gusta hacer sufrir a los demás	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_7	La paso mal en la sala de clase	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
g1_8	A mis compañeros(as) les gusta poner sobrenombres	1 = Nunca; 2 = Pocas veces; 3 = Casi siempre; 4 = Siempre
<i>Panel B. Cyber-victimization, last 12 months (any item)</i>		
g2_1	¿En los últimos 12 meses has recibido mensajes ofensivos por medio de WhatsApp u otros servicios de mensajería?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_2	¿En los últimos 12 meses han publicado fotografías o videos vergonzosos de ti, a través de redes sociales?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_3	¿En los últimos 12 meses has recibido llamadas al celular insultantes o amenazantes?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_4	¿En los últimos 12 meses has recibido mensajes de correo electrónico insultantes o amenazantes?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_5	¿En los últimos 12 meses has leído o tenido conocimiento de conversaciones en grupos de chat en las que se expresan frases ofensivas sobre ti?	1 = Sí; 2 = No; 99 = Prefiero no responder
g2_6	¿En los últimos 12 meses has visitado o tenido conocimiento sobre páginas web donde se burlan de ti, o se descarga información personal sin tu permiso?	1 = Sí; 2 = No; 99 = Prefiero no responder
<i>Panel C. Dating violence, among adolescents who ever dated (any item)</i>		
g9_1	Violencia física	0 = No; 1 = Sí
g9_2	Violencia psicológica	0 = No; 1 = Sí
g9_3	Violencia sexual	0 = No; 1 = Sí
<i>Panel D. Substance use, last 12 months</i>		
g11	¿Has fumado cigarrillos de tabaco alguna vez en los últimos 12 meses?	1 = Sí; 2 = No
g12_1	Cerveza	1 = Sí; 2 = No
g12_2	Vino	1 = Sí; 2 = No
g12_3	Licores solos o mezclados con refrescos (pisco, vodka, ron, gin)	1 = Sí; 2 = No
g12_4	Cualquier otra bebida que contenga alcohol	1 = Sí; 2 = No
g13_1	Cannabis (Marihuana, hachís, creepy)	1 = Sí; 2 = No

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. Drank alcohol is one if any of the four beverage items is answered yes; cyber-victimization is one if any of the listed items is answered yes.

TABLE A4—MATERNAL MENTAL-HEALTH QUESTIONS: ITEMS AND SCALES

	Question, verbatim (1)	Response scale (2)
<i>Panel A. Reported by the mother in 2010, about the pregnancy and postpartum of the sampled child</i>		
g4a_1	Depresión. ¿Durante el embarazo le diagnosticaron alguna de las siguientes...	Marked = yes; not marked = no
g4a_3	Trastorno de ansiedad generalizada. ¿Durante el embarazo le diagnosticaron...	Marked = yes; not marked = no
g4a_7	Trastorno estrés postraumático. ¿Durante el embarazo le diagnosticaron a...	Marked = yes; not marked = no
g4b	¿Tuvo alguna derivación psicológica o psiquiátrica?	1 = sí; 2 = no; 9 = no sabe no responde
g19	Posterior al embarazo, ¿se le diagnosticó por un especialista, depresión po...	1 = sí; 2 = no; 9 = no sabe
<i>Panel B. Reported by the caregiver in 2012</i>		
b64	Últimamente, algún especialista, ¿le ha diagnosticado a Ud. depresión?	1 = sí; 2 = no; 8 = no responde
b55o_madre	Depresión. Le mencionaré una lista de enfermedades, ¿me podrí...	1 = sí; 2 = no; 9 = no sabe
b55p_madre	Ansiedad. Le mencionaré una lista de enfermedades, ¿me podría...	1 = sí; 2 = no; 9 = no sabe

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. A mother is vulnerable in Table 12 if any Panel A item is answered yes. The 2012 items enter the family mental-health controls of the main specification.

TABLE A5—PLACEBO OUTCOMES FIXED AT BIRTH: ITEMS AND SCALES

	Question, verbatim (1)	Response scale (2)
g24	¿Cuál fue el peso del(de la) niño(a) al nacer?	Open numeric answer (kilograms); special codes: 88 = dato impugnado; 99 = no sabe no responde
g23	¿Cuál fue la talla, altura del niño(a) al nacer?	Open numeric answer (centimeters); special codes: 88 = dato impugnado; 99 = no sabe no responde
g20	¿Cuántas semanas de embarazo tenía Ud. cuando nació el(la) niño(a)?	Open numeric answer (weeks); special codes: 88 = dato impugnado; 99 = no sabe no responde
g17_6	Parto prematuro (antes de las 37 semanas). Al momento del nacimiento, ¿ex...	1 = sí; 2 = no; 9 = no sabe no responde
g25b1	b1)Medición 2 (5 Minutos). ¿Cuál fue su puntuación en el Test de Apgar?	Open numeric answer (score 0–10); special codes: 99 = no sabe no responde

Notes: Question wording and response labels are reproduced verbatim in the original Spanish of the questionnaire, from the variable and value labels of the public microdata; wording the microdata truncates at 80 characters is marked with an ellipsis. Cleaning: birth weight kept between 0.5 and 6.5 kilograms, length between 30 and 65 centimeters, gestation between 24 and 45 weeks, Apgar between 0 and 10; the survey codes 99 for a forgotten answer, treated as missing. All five outcomes are fixed at birth, one to four years before the earthquake for every cohort of the analysis sample.